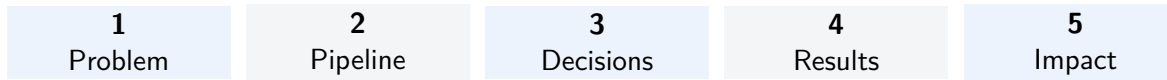




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- ▶ Problem and objective
- ▶ Our 6-stage pipeline
- ▶ Key engineering decisions
- ▶ Results and evaluation
- ▶ Practical impact, limitations, and conclusion

Problem

Manual workflow today

- ▶ Signature writing is slow
- ▶ Inconsistent quality between analysts
- ▶ Hard to scale to large malware volumes
- ▶ Repetitive effort for similar families

Target workflow

- ▶ AI-assisted signature drafting
- ▶ Faster analyst turnaround
- ▶ Standardized detection logic
- ▶ Human validation before deployment

Core question: Can LLMs generate useful CAPEv2 signatures with strong precision and practical analyst value?

- ▶ Dataset: 100 malware samples across 10 families
- ▶ Input: CAPEv2 dynamic analysis JSON reports
- ▶ Output: Python CAPEv2 signatures + MITRE ATT&CK mappings
- ▶ Constraint: Low-cost approach using local Llama inference

Deliverables

6 notebooks, processed outputs, 20 generated signatures, evaluation report, and containerized setup.

Approach

1. Data Acquisition



2. Feature Extraction



3. Behavior I

4. LLM Analysis



5. Signature Generation



6. Evaluation

Design principle

Separate data normalization from LLM reasoning to improve reliability and reproducibility.

Extracted behavior groups

- ▶ Process and injection activity
- ▶ Registry persistence indicators
- ▶ File operations and drops
- ▶ Network and command behavior

Behavior IR value

- ▶ Normalizes inconsistent report structures
- ▶ Reduces prompt noise for the LLM
- ▶ Improves reproducibility across runs

- ▶ Llama used for semantic interpretation (local, low cost)
- ▶ Family-level aggregation to detect recurring patterns
- ▶ CAPEv2 signature generation with strict validation:
 - syntax checks
 - logic checks
 - manual analyst review

Important result: Validation caught approximately 15% of generated signature issues before final output.

Engineering Reality

Challenge

How we solved it

Inconsistent JSON structures

Built robust parsing + Behavior IR abstraction

LLM output quality drift

Added cache, tighter prompts, post-checks

Signature correctness gaps

Added syntax + logic + manual validation

Lower ATT&CK edge-case accuracy

Added confidence-aware review workflow

- ▶ Feeding raw CAPEv2 JSON directly to Llama
- ▶ Fully automated ATT&CK mapping without confidence checks
- ▶ Treating generated Python as deployment-ready without validation

Key lesson

Automation in security must include explicit validation gates. Speed without verification creates detection risk.

Results

Precision	Recall	F1-score
86%	79%	0.82

Quality Details	Output Volume
▶ False positive rate: 8% ▶ Validation catches edge-case bugs ▶ Human review remains in the loop	▶ 20 CAPEv2 signatures ▶ 10 malware families covered ▶ 35+ ATT&CK techniques mapped

- ▶ Cross-family precision on unseen families: 77%
- ▶ Cross-family recall on unseen families: 65%
- ▶ ATT&CK mapping overall accuracy: 84%

Interpretation

The model captures reusable behavior patterns, but still requires analyst oversight for edge cases such as subtle Discovery and Lateral Movement indicators.

- ▶ Practical analyst acceleration: from hours to minutes for initial signature drafts
- ▶ Repeatable pipeline with containerized setup
- ▶ Cost-aware execution using local inference
- ▶ Strong baseline for SOC workflow augmentation

Bottom line: this is an analyst multiplier, not a full analyst replacement.

Conclusion

- ① LLM-assisted CAPEv2 signature generation is feasible and useful.
- ② A Behavior IR layer is critical for stability and quality.
- ③ Validation is mandatory for safe security automation.
- ④ The system provides immediate practical value with clear known limits.

- ▶ Expand dataset size and family diversity
- ▶ Improve low-confidence ATT&CK mapping with hybrid rules
- ▶ Add automated regression tests for signatures
- ▶ Integrate with SOC tooling for continuous feedback

Thank you

Questions?

Repository: <https://github.com/ykram051/Automated-Dynamic-Analysis-Signature-Generation>